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Title

**Noise-Aware Geometry-Guided Graph
Neural Initialization for Variational
Quantum Eigensolvers**

REVIEW REPORT – I

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Abstract

Variational Quantum Eigensolvers (VQE) are among the most promising near-term algorithms for quantum chemistry and condensed-matter simulation on noisy intermediate-scale quantum (NISQ) hardware, but their practical performance is strongly limited by poor parameter initialization, barren plateaus, and the high cost of quantum-hardware-aware optimizers such as Quantum Natural Gradient Descent (QNGD). Existing Graph Neural Network (GNN) based initializers for VQE map a single problem/ansatz graph to an initial parameter vector but do not explicitly and jointly condition on hardware topology and device noise, and do not attempt to learn the local optimization geometry used by QNGD. This project proposes a dual-graph, noise-aware neural architecture that jointly encodes the problem Hamiltonian, the parameterized ansatz, and the hardware topology together with its noise profile, and predicts both a high-quality initial parameter vector and a local approximation to the quantum-geometric metric used by QNGD. The predicted initialization is evaluated in a VQE loop and adaptively refined using geometry-aware updates, with the aim of reducing the number of expensive quantum circuit evaluations required to reach a target energy. Two complementary dataset strategies are defined: an existing-dataset strategy built around the VQEzy VQE-initialization dataset for rapid baseline implementation, and a self-generated dataset strategy that provides full control over Hamiltonian family, ansatz depth, hardware topology, and noise for the final noise-robustness and generalization study. A systematic ablation plan isolates the contribution of dual-graph encoding, cross-attention fusion, noise conditioning, the geometry head, and adaptive refinement, complemented by an auxiliary ablation study on the Quantum Approximate Optimization Algorithm (QAOA) that separates simple noise-aware feature engineering from the proposed architectural and algorithmic contribution. The overall contribution integrates problem structure, circuit structure, hardware/noise awareness, parameter initialization, and geometry-aware optimization within a single VQE framework.

Keywords: Variational Quantum Eigensolver; Graph Neural Network; Quantum Natural Gradient Descent; Noise-Aware Initialization; Parameterized Quantum Circuits

1. Introduction

1.1 Background

The Variational Quantum Eigensolver is a hybrid quantum-classical algorithm that estimates the ground-state energy of a Hamiltonian H by preparing a parameterized quantum state $|\psi(\vartheta)\rangle = U(\vartheta)|\psi_0\rangle$ on quantum hardware and minimizing the expectation value

$$E(\vartheta) = \langle\psi(\vartheta)|H|\psi(\vartheta)\rangle \quad (1.1)$$

using a classical optimizer. Quantum Natural Gradient Descent improves on plain gradient descent by preconditioning the update with the quantum-geometric (Fisher information) metric $F(\vartheta)$,

$$\vartheta^{(t+1)} = \vartheta^{(t)} - \eta F(\vartheta^{(t)})^{-1} \nabla E(\vartheta^{(t)}), \quad (1.2)$$

but both the optimization landscape and the achievable accuracy are highly sensitive to the starting point ϑ_0 , to the ansatz structure, and to the hardware topology and noise on which the circuit is executed.

1.2 Motivation

Parameter initialization is one of the central practical bottlenecks in VQE: a poor starting point leads to slow convergence, susceptibility to barren plateaus, and wasted quantum circuit evaluations, which are the most expensive resource on NISQ hardware. Existing GNN-based initializers for VQE, such as Qracle, map a single problem/ansatz graph to a parameter vector but do not explicitly and jointly condition on hardware topology and device noise, and do not attempt to learn the local optimization geometry used by QNGD. The motivation for this work is to integrate hardware topology, device noise, and geometry-aware optimization into one coherent, noise-aware initialization-and-refinement framework for VQE, and to demonstrate through controlled ablations that each additional component contributes measurable value beyond existing GNN initializers and standard QNGD.

1.3 Objectives

1. To design a dual-graph, noise-aware neural architecture that jointly encodes the Hamiltonian graph, the ansatz/circuit graph, and the hardware-topology/noise

graph for VQE parameter initialization.

2. To develop a joint parameter-and-geometry prediction model $f_\theta(G_H, G_A, G_D, N) \rightarrow (\vartheta_0, \hat{F}_0)$ that supplies both an initial parameter vector and a local approximation of the quantum-geometric metric used by QNGD.
3. To implement an adaptive QNGD refinement procedure that uses the predicted initialization and geometry to reduce the number of quantum circuit evaluations required to reach a target energy threshold.
4. To define and compare two complementary dataset strategies – an existing-dataset strategy (VQEzy and related public VQE datasets) and a self-generated, fully controllable dataset – for training and evaluating the proposed model.
5. To evaluate the proposed method against random, heuristic, and existing GNN-based (Qracle-style) initialization baselines, and to conduct a systematic ablation study isolating the contribution of dual-graph encoding, cross-attention fusion, noise conditioning, the geometry head, and adaptive refinement.

1.4 Scope

The scope of the present work is restricted to parameter initialization and geometry-aware refinement for VQE on Ising-type, Heisenberg-type, and small chemistry-inspired Hamiltonians, evaluated primarily through ideal and noisy quantum simulation, with limited validation on real NISQ hardware where resources permit. The project does not attempt to redesign QNGD itself. A QAOA-based ablation study is included solely to separate the contribution of the proposed noise-aware and geometry-aware components from simple feature-level noise conditioning; QAOA is not the primary target application of this work, and all architectural and algorithmic contributions are developed and evaluated in the VQE setting.

2. Related Work

Table 2.1 summarizes the key related work in GNN-based parameter initialization for VQE and in the datasets available for this line of research. Two closely related GNN-based initializers for the Quantum Approximate Optimization Algorithm (QAOA) are also included, since the noise-aware and geometry-aware components proposed for VQE in this project are additionally evaluated on QAOA as a controlled ablation setting (Chapter 5).

Table 2.1: Summary of related work

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Zhang et al., Qracle (2025)[3]	GNN-based parameter initializer for VQE.	Demonstrates that a GNN can map problem/ansatz structure to a VQE initial parameter vector.	Maps a single problem/ansatz graph to parameters only; no explicit hardware/noise graph or geometry prediction.
2	Cheng et al. (2025)[4]	Clifford-circuit-based initialization for variational quantum algorithms.	Provides a non-learned, classically simulable initialization strategy competitive with random/heuristic starts.	Does not use learned structure from the Hamiltonian/ansatz/hardware; limited adaptivity to noise.
3	VQezy dataset (2025)[5]	Public dataset of 12,110 VQE instances across three domains and seven tasks with full optimization trajectories.	Provides ready-made trajectories suitable for benchmarking learned initialization and convergence.	Exact hardware/noise fields, licensing, and task coverage require verification before use.
4	Jain et al. (2022)[1]	GNN-based warm-starting of QAOA parameters from the problem graph.	Established that GNNs can generalize parameter initialization across graph instances and sizes.	Restricted to QAOA; no hardware or noise conditioning; angle prediction only.
5	Jiang et al., QSeer (2025)[2]	Quantum-inspired physics-informed GNN for QAOA parameter initialization.	Improves on plain GNN warm-starting by embedding physics-informed structure.	QAOA-specific; does not jointly model hardware noise or optimizer geometry.

2.1 Research Gap

Existing GNN-based initializers for VQE, such as Qracle, learn a mapping from a single problem/ansatz graph to a parameter vector, and Clifford-based initialization avoids learning altogether. None of these approaches jointly :

- (i) represent the Hamiltonian, ansatz, and hardware/noise structure as separate, fused graph representations
- (ii) predict both an initial parameter vector and a local approximation of the quantum-geometric metric used by QNGD, and
- (iii) couple that joint prediction with an adaptive, geometry-aware refinement loop designed to reduce quantum circuit evaluations.

This combination hardware and noise awareness, joint parameter-and-geometry prediction, and adaptive QNGD refinement, constitutes the research gap addressed by this project.

3. Problem statement

Given a Hamiltonian H , a parameterized ansatz $U(\vartheta)$ with p parameters, a hardware topology D , and an estimated noise profile N , the problem is to learn a function

$$f_{\theta}(G_H, G_A, G_D, N) \rightarrow (\vartheta_0, \hat{F}_0) \quad (3.1)$$

where G_H , G_A , and G_D are graph representations of the Hamiltonian, the ansatz/circuit, and the hardware topology respectively, ϑ_0 is a predicted initial parameter vector for the VQE objective in Eq. (1.1), and $\hat{F}_0$ is a learned local approximation of the quantum-geometric metric used by QNGD in Eq. (1.2). The initialization ϑ_0 is subject to the constraint that it must be produced without requiring the expensive iterative VQE optimization to be run to completion for the target instance, and the model must generalize to unseen Hamiltonians, unseen ansatz depths, unseen qubit counts, and unseen noise levels not present during training. Success is measured against baselines by :

- (i) the initial energy $E(\vartheta_0)$
- (ii) the final energy error after a fixed evaluation budget
- (iii) the number of quantum circuit evaluations required to reach a target energy threshold
- (iv) the degradation of these quantities under increasing device noise.

4. Justification for the problem formulation

4.1 Need for the Proposed Work

Parameter initialization is one of the main practical bottlenecks in VQE: poor starting points lead to slow convergence, susceptibility to barren plateaus, and wasted quantum circuit evaluations, which are the most expensive resource on NISQ hardware. Existing GNN-based initializers for VQE, such as Qracle, do not explicitly and jointly condition on hardware topology and device noise, and do not attempt to jointly learn the local optimization geometry used by QNGD. Addressing this gap directly targets a resource (circuit evaluations) whose cost is the primary practical constraint of the NISQ era.

4.2 Feasibility

Technical feasibility: the required components – graph neural network encoders, cross-attention fusion, and QNGD implementations – are supported by existing quantum-simulation and machine-learning libraries (e.g., PennyLane-based simulators together with standard GNN toolkits), and the project can be developed incrementally, first on classical simulation and later, where resources allow, validated on small NISQ instances. *Data feasibility:* two complementary dataset strategies are available – the existing VQEzy dataset for a fast initial implementation, and a self-generated dataset for full control over hardware and noise variables – so the project is not blocked on a single data source. *Schedule feasibility:* the recommended development sequence (Table 6.1 in Chapter 6) proceeds from a VQE+QNGD baseline, through the incremental addition of dual-graph encoding, noise conditioning, the geometry head, and adaptive refinement, to a final generalization study and evaluation, with the QAOA ablation study run alongside as an auxiliary check on the noise-aware and geometry-aware components.

4.3 Expected Contribution

The project is expected to contribute: (i) a dual-graph representation linking Hamiltonian/optimization structure with ansatz and hardware structure; (ii) a noise-conditioned message-passing and fusion mechanism for hardware-aware VQE initialization; (iii) a parameter-and-geometry prediction architecture that supplies both a starting point and useful local optimization information; (iv) a label-efficient, quantum-in-the-loop

training strategy that reduces dependence on expensive pre-optimized parameter labels; (v) an adaptive QNGD refinement strategy aimed at reducing quantum circuit evaluations; and (vi) a systematic ablation framework that separates architectural, optimization, and noise-robustness contributions. The ablation framework in Chapter 5 is designed to demonstrate that each of these components provides a measurable improvement over a plain VQE+QNGD baseline and over existing GNN-based VQE initializers (Table 2.1).

5. Methodology

The proposed system follows a modular workflow comprising dataset construction, dual-graph problem/ansatz/hardware representation, joint parameter-and-geometry prediction, quantum evaluation, and adaptive QNGD-based refinement, followed by systematic ablation and evaluation.

5.1 Architectural Design of the Proposed System

The proposed architecture consists of the **Hamiltonian Encoder**, the **Ansatz Encoder**, the **Hardware/Noise Encoder**, a **Cross-Attention Fusion** stage, a **Layer-wise Parameter Decoder**, a **Geometry Head**, a **Quantum Evaluation Module**, and an **Adaptive QNGD Module**, as illustrated in Figure 5.1.

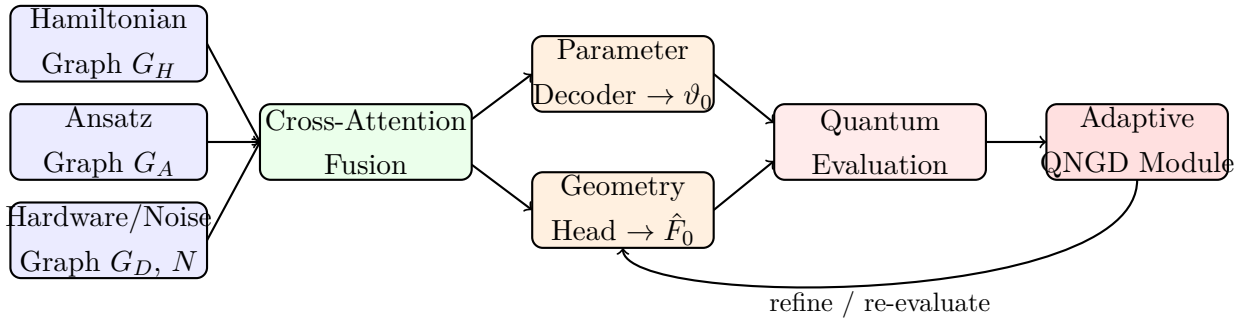


Figure 5.1: Proposed dual-graph, noise-aware, geometry-guided VQE initialization and refinement architecture

The **Hamiltonian Encoder** learns interaction structure from the operator/Hamiltonian graph, whose node and edge features include qubit nodes, Pauli/operator type, coefficient magnitude, interaction order, and weighted edges. The **Ansatz Encoder** learns parameterized-circuit structure and layer dependencies from gate type, parameter index, circuit layer, gate connectivity, and parameter-sharing features. The **Hardware/Noise Encoder** represents physical-qubit connectivity, coupling edges, gate availability, and noise descriptors such as two-qubit gate error, one-qubit gate error, readout error, and coherence-related features. **Cross-Attention Fusion** learns which aspects of the problem structure and the implementation structure should influence the variational initialization, with noise injected into message computation or feature conditioning rather than appended only at the final layer.

The **Layer-wise Parameter Decoder** predicts the initial parameter vector

$$\vartheta_0 = [\vartheta_1, \vartheta_2, \dots, \vartheta_p], \quad (5.1)$$

optionally in a hierarchical form so that layer l receives the shared problem/hardware representation together with a learned layer embedding. The **Geometry Head** predicts a representation of the local metric/preconditioner $\hat{F}_0$ for use in the QNGD update of Eq. (1.2); a conservative implementation first uses $\hat{F}_0$ as a warm-start preconditioner and compares it against an exact/estimated metric before it is trusted as a full replacement. The **Quantum Evaluation Module** evaluates the VQE energy $E(\vartheta_0)$ under ideal and/or noisy simulation, and the **Adaptive QNGD Module** performs geometry-aware refinement, terminating early when the initialization already meets a quality threshold.

5.1.1 Training objective

The model can be trained using reference-optimized parameter labels where available, with a parameter-prediction loss

$$\mathcal{L}_{param} = \|\vartheta_0 - \vartheta^*\|_2^2, \quad (5.2)$$

and, in the label-efficient / quantum-in-the-loop setting, a combined objective evaluated directly against the quantum simulator,

$$\mathcal{L} = E_{noise}(\vartheta_0) + \lambda_1 \mathcal{L}_{geometry} + \lambda_2 \mathcal{L}_{robustness}, \quad (5.3)$$

where $E_{noise}(\vartheta_0)$ encourages a useful initialization under sampled noise, $\mathcal{L}_{geometry}$ enforces consistency between the predicted metric $\hat{F}_0$ and the measured/estimated local metric, and $\mathcal{L}_{robustness}$ encourages stability across noise conditions. The exact coefficients λ_1, λ_2 must be tuned experimentally.

5.2 Modules

5.2.1 Module 1: Dataset and Input Management

Inputs

- Molecular and synthetic Hamiltonians (Ising, Heisenberg, and chemistry-oriented families from PennyLane quantum-chemistry datasets).
- Ansatz configurations across multiple circuit depths (initially 1–5).
- Hardware topologies (linear, ring, grid, and device-inspired connectivity) and noise conditions (low/medium/high or continuous parameter ranges).

- Existing VQE datasets: VQEzy[5], the VQE-generated Quantum Circuit Dataset (Qulacs/Osaka)[6], and the SIAM VQE dataset from Oak Ridge National Laboratory[7].

Processing Two complementary dataset strategies are used together rather than in isolation. The first strategy uses **VQEzy**, which reports 12,110 instances across three domains and seven representative tasks with complete optimization trajectories, as the fastest route to an initial implementation; the VQE-generated Quantum Circuit Dataset (covering 4, 8, 12, 16, and 20 qubits across multiple Hamiltonian/ansatz families) and the SIAM VQE dataset (Anderson impurity models with ideal and sampling-noise-related results) are used as independent complementary benchmarks. Where an existing dataset lacks the required hardware/noise attributes, it is augmented through a documented mapping – existing VQE record $\rightarrow$ hardware mapping $\rightarrow$ controlled noise simulation $\rightarrow$ augmented graph instance – with the original source cited and the augmentation procedure released for reproducibility. The second strategy involves the usage of a self-generated dataset systematically varying Hamiltonian family, qubit count, ansatz depth, hardware topology, and noise level to produce instances of the form (Hamiltonian H , Ansatz A , Hardware topology D , Noise profile N , reference/trajectory information, evaluation energy), stored as JSON, NumPy/PyTorch tensors, or a relational table with linked graph objects; this plan is used for the final noise-robustness and generalization study, where full control over hardware and noise is required. Data is split so that validation and test sets specifically probe unseen Hamiltonians/geometries, unseen circuit depths, unseen qubit counts, and unseen noise levels or hardware profiles, rather than relying on a random row-level split.

Outputs

- Hamiltonian, ansatz, and hardware/noise graph representations for each instance.
- Reference parameters and optimization trajectories, where available.
- Ideal and noisy simulation results.
- Training/validation/test splits designed for generalization testing (unseen Hamiltonians, depths, qubit counts, and noise levels).

Dependencies VQEzy and related public VQE datasets, the PennyLane quantum-chemistry data sources, and the VQE/QNGD simulation framework used for Plan A generation and for augmenting Plan B instances.

5.2.2 Module 2: Core Processing – Dual-Graph Encoding and Joint Prediction

The Hamiltonian, ansatz, and hardware/noise graphs are encoded by dedicated GNN encoders and fused through cross-attention and noise-conditioned message passing, as described in Section 5.1. For an instance (G_H, G_A, G_D, N) , the model learns the joint mapping of Eq. (3.1), producing the initial parameter vector ϑ_0 (Eq. (5.1)) and the geometry approximation $\hat{F}_0$. Training combines the parameter-prediction loss (Eq. (5.2)) with the quantum-in-the-loop objective (Eq. (5.3)) so that the model is not entirely dependent on pre-optimized labels.

Outputs

- Predicted initial parameter vector ϑ_0 .
- Predicted geometry representation $\hat{F}_0$.
- Learned dual-graph representations.
- Trained model parameters.

Dependencies The GNN framework, the VQE/QNGD simulator, and Module 1 outputs.

5.2.3 Module 3: Quantum Evaluation and Adaptive QNGD Refinement

This module implements Algorithm 1. It evaluates the VQE energy $E(\vartheta_0)$ under ideal and/or noisy simulation, decides whether the predicted initialization already meets a quality threshold, and, if not, performs QNGD updates using the measured/estimated metric, optionally initialized or preconditioned by $\hat{F}_0$.

Algorithm 1 Noise-Aware Geometry-Guided VQE Initialization and Refinement

1. Construct the Hamiltonian graph G_H from the target Hamiltonian.
 2. Construct the ansatz/circuit graph G_A from the chosen parameterized circuit.
 3. Construct the hardware graph G_D and collect an estimated noise profile N .
 4. Encode G_H , G_A , and G_D using dedicated graph encoders.
 5. Fuse the representations using cross-attention and noise-conditioned message passing.
 6. Predict the initial parameter vector ϑ_0 and geometry representation $\hat{F}_0$.
 7. Evaluate the VQE energy $E(\vartheta_0)$ under the selected simulator/hardware noise model.
 8. If the predicted initialization satisfies the quality threshold, terminate or use a small refinement budget.
 9. Otherwise perform QNGD updates using the measured/estimated metric, optionally initialized or preconditioned by $\hat{F}_0$.
 10. Record convergence iterations, circuit evaluations, final energy, energy error, gradient norm, and robustness metrics.
-

Outputs

- Initial and final VQE energies and energy error.
- Number of QNGD iterations and quantum circuit evaluations.
- Gradient-norm and robustness measurements.

Dependencies Module 2 outputs and the VQE/QNGD simulation and hardware-execution infrastructure.

5.2.4 Module 4: Testing and Evaluation

The proposed model is compared against random VQE initialization with gradient descent, random VQE initialization with QNGD, standard heuristic/physics-based initialization with QNGD where applicable, Qracle-style GNN initialization, the proposed GNN initialization with standard QNGD, and the full proposed joint parameter-

and-geometry model with adaptive QNGD, each under ideal and noisy execution. Primary evaluation metrics are the initial energy, final energy/energy error, iterations to threshold, quantum circuit evaluations, gradient norm/trainability, noise-induced energy degradation, generalization error on unseen Hamiltonians/ansätze/depths/noise, and wall-clock time.

An ablation study isolates each claimed contribution: B0 removes all proposed components (basic VQE+QNGD baseline); B1 removes the Hamiltonian/ansatz dual encoding; B2 removes cross-attention; B3 removes noise conditioning; B4 removes the geometry head; B5 removes adaptive refinement; and B6 is the full proposed model. The critical comparisons are the full model against a Qracle-style GNN (to establish architectural contribution beyond GNN VQE initialization), against parameter-only prediction (to isolate the geometry head), against noise-feature concatenation (to isolate true noise-conditioned message passing), against standard QNGD (to measure whether predicted geometry improves the optimizer), and ideal versus noisy execution (to demonstrate whether the hardware-aware design is meaningful). An auxiliary QAOA-based ablation study (A0–A5, comparing random, QSeer-style, noise-aware-feature-only, and the proposed joint architecture applied to QAOA parameter initialization) is additionally used to separate the contribution of simple noise-aware feature engineering from the deeper architectural and algorithmic contribution that the proposed method targets for VQE.

Inputs

- Trained models (full and ablated variants).
- Baseline initialization methods.
- Test instances spanning unseen Hamiltonians, depths, qubit counts, and noise levels.
- Noise models and, where available, real-hardware execution results.

Outputs

- Quantitative comparison against baseline and existing GNN-based initializers.
- Ablation results isolating architectural, optimization, and noise-robustness contributions.
- Generalization results across unseen problem sizes, depths, and noise conditions.

Dependencies Outputs of Modules 1–3, the VQE/QNGD simulation infrastructure, and available quantum hardware for small-scale validation.

6. Project timeline chart

The recommended development sequence proceeds in eight phases, each gated by a specific decision point before advancing, as summarized in Table 6.1 and Figure 6.1.

Table 6.1: Recommended development sequence and decision gates

Phase	Work	Decision gate
Phase 1	Implement VQE + QNGD baseline.	Establish reliable energy and convergence measurements.
Phase 2	Implement dual-graph GNN parameter initializer.	Test architecture without the geometry head.
Phase 3	Add noise-conditioned message passing.	Verify improvement under controlled noise.
Phase 4	Add geometry head.	Test whether predicted metric/preconditioner adds value.
Phase 5	Add adaptive refinement.	Measure circuit-evaluation savings.
Phase 6	Run the auxiliary QAOA ablation study (A0–A5).	Confirm that noise-aware and geometry-aware components add value beyond simple feature engineering.
Phase 7	Cross-instance/depth/noise generalization study.	Test transfer rather than only interpolation.
Phase 8	Final evaluation and report/paper write-up.	Consolidate results into the final manuscript.

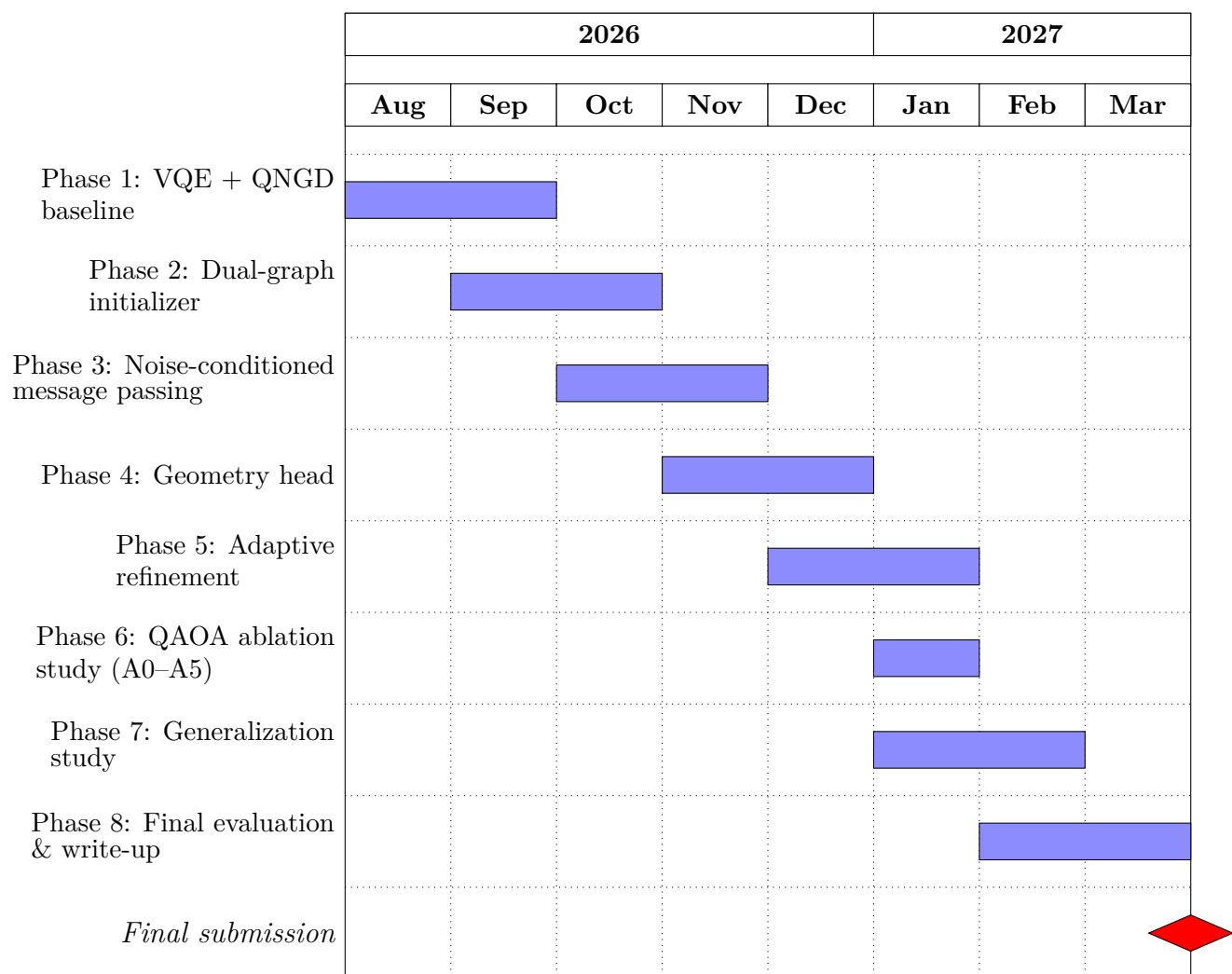


Figure 6.1: Project timeline

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